

Time Series — Solutions

Practice Exam — Week 3: Backshift, Wold Decomposition, Causality & Invertibility
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Problem 1 (5 points)

(a) The *backshift operator* B shifts a series one step into the past, $BX_t = X_{t-1}$, and iterating $B^k X_t = X_{t-k}$ (with $B^0 = I$). For an ARMA(p, q) written $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the characteristic polynomials are

$$\Phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p, \quad \Theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$$

(coefficients matching the chosen sign convention). The root criteria are:

- **Stationarity / causality:** all roots of $\Phi(z)$ lie *outside* the unit circle, $|z| > 1$;
- **Invertibility:** all roots of $\Theta(z)$ lie *outside* the unit circle, $|z| > 1$.

(b) In backshift form $(1 - \frac{2}{3}B + \frac{1}{6}B^2)X_t = \varepsilon_t$, so

$$\Phi(z) = 1 - \frac{2}{3}z + \frac{1}{6}z^2.$$

Solve $\Phi(z) = 0$. Multiplying by 6 gives $z^2 - 4z + 6 = 0$, hence

$$z = \frac{4 \pm \sqrt{16 - 24}}{2} = 2 \pm \sqrt{-2} = 2 \pm \sqrt{2}i, \quad |z| = \sqrt{2^2 + (\sqrt{2})^2} = \sqrt{4 + 2} = \sqrt{6} \approx 2.449.$$

Both roots have modulus $\sqrt{6} > 1$, i.e. lie outside the unit circle, so the process is **stationary** (causal). (One may also check the AR(2) triangle with $\phi_1 = \frac{2}{3}$, $\phi_2 = -\frac{1}{6}$: $\phi_1 + \phi_2 = \frac{1}{2} < 1$, $\phi_2 - \phi_1 = -\frac{5}{6} < 1$, $|\phi_2| = \frac{1}{6} < 1$.)

(c) **Yes, invertible.** It is a pure AR process ($\Theta \equiv 1$, a finite polynomial with no roots to test), so $\varepsilon_t = \Phi(B)X_t$ is already a finite expression in present and past X : every AR is automatically invertible.

Problem 2 (5 points)

(a) $\Phi(z) = 1 - 0.7z + 0.1z^2 = 0.1(z^2 - 7z + 10) = 0.1(z - 2)(z - 5)$, so the AR roots are $z = 2$ and $z = 5$. Both satisfy $|z| > 1$, hence all roots of Φ are outside the unit circle and the process is **causal** (stationary). Also $\Theta(z) = 1 + 0.5z$ has root $z = -2$, $|z| = 2 > 1$, so it is invertible as well (not asked, but it guarantees a well-defined G).

(b) Write $X_t = G(B)\varepsilon_t$ with

$$G(z) = \frac{1 + 0.5z}{0.1(z - 2)(z - 5)}.$$

Partial-fraction the rational factor:

$$\frac{1}{(z - 2)(z - 5)} = \frac{A}{z - 2} + \frac{B}{z - 5}, \quad A = \frac{1}{2 - 5} = -\frac{1}{3}, \quad B = \frac{1}{5 - 2} = \frac{1}{3}.$$

Convert each pole to a power series convergent on (and inside) the unit disk, using $|z| < 2 < 5$:

$$\begin{aligned} \frac{-1}{z - 2} &= \frac{1}{2} \frac{1}{1 - z/2} = \frac{1}{2} \sum_{j \geq 0} \left(\frac{1}{2}\right)^j z^j = \sum_{j \geq 0} \left(\frac{1}{2}\right)^{j+1} z^j, \\ \frac{1}{z - 5} &= -\frac{1}{5} \frac{1}{1 - z/5} = -\frac{1}{5} \sum_{j \geq 0} \left(\frac{1}{5}\right)^j z^j = -\sum_{j \geq 0} \left(\frac{1}{5}\right)^{j+1} z^j. \end{aligned}$$

Hence, collecting $\frac{1}{(z-2)(z-5)} = \frac{A(-1)}{2-z} \dots$ explicitly,

$$\frac{1}{(z-2)(z-5)} = \left(-\frac{1}{3}\right) \frac{1}{z-2} + \frac{1}{3} \frac{1}{z-5} = \sum_{j \geq 0} \left[\frac{1}{3} \left(\frac{1}{2}\right)^{j+1} - \frac{1}{3} \left(\frac{1}{5}\right)^{j+1} \right] z^j.$$

Now multiply by $\frac{1+0.5z}{0.1} = 10(1+\frac{1}{2}z)$. Write $H(z) = \sum_{j \geq 0} h_j z^j$ with

$$h_j = 10 \left[\frac{1}{3} \left(\frac{1}{2}\right)^{j+1} - \frac{1}{3} \left(\frac{1}{5}\right)^{j+1} \right] = \frac{5}{3} \left(\frac{1}{2}\right)^j - \frac{2}{3} \left(\frac{1}{5}\right)^j,$$

so that $G(z) = (1+\frac{1}{2}z)H(z)$ gives $\psi_0 = h_0$ and $\psi_j = h_j + \frac{1}{2}h_{j-1}$ for $j \geq 1$. Computing the coefficients of $(1/2)^j$ and $(1/5)^j$:

$$\begin{aligned} \text{coeff of } \left(\frac{1}{2}\right)^j : & \quad \frac{5}{3} + \frac{1}{2} \cdot \frac{5}{3} \cdot 2 = \frac{5}{3} + \frac{5}{3} = \frac{10}{3}, \\ \text{coeff of } \left(\frac{1}{5}\right)^j : & \quad -\frac{2}{3} + \frac{1}{2} \cdot \left(-\frac{2}{3}\right) \cdot 5 = -\frac{2}{3} - \frac{5}{3} = -\frac{7}{3}, \end{aligned}$$

(using $(\frac{1}{2})^{j-1} = 2(\frac{1}{2})^j$ and $(\frac{1}{5})^{j-1} = 5(\frac{1}{5})^j$). Therefore

$$\boxed{\psi_j = \frac{10}{3} \left(\frac{1}{2}\right)^j - \frac{7}{3} \left(\frac{1}{5}\right)^j, \quad j \geq 0}$$

which already gives $\psi_0 = \frac{10}{3} - \frac{7}{3} = 1$. In particular

$$\psi_0 = 1, \quad \psi_1 = \frac{10}{3} \cdot \frac{1}{2} - \frac{7}{3} \cdot \frac{1}{5} = \frac{5}{3} - \frac{7}{15} = \frac{6}{5} = 1.2, \quad \psi_2 = \frac{10}{3} \cdot \frac{1}{4} - \frac{7}{3} \cdot \frac{1}{25} = \frac{5}{6} - \frac{7}{75} = \frac{37}{50} = 0.74.$$

Check (coefficient matching). From $(1-0.7z+0.1z^2)\sum_j \psi_j z^j = 1+0.5z$: $\psi_0 = 1$; $\psi_1 - 0.7\psi_0 = 0.5 \Rightarrow \psi_1 = 1.2$; and $\psi_j = 0.7\psi_{j-1} - 0.1\psi_{j-2}$ for $j \geq 2$, giving $\psi_2 = 0.7(1.2) - 0.1(1) = 0.74$, matching exactly.

Problem 3 (5 points)

(a) The model is $X_t - \frac{5}{6}X_{t-1} + \frac{1}{6}X_{t-2} = \varepsilon_t$, with $\Phi(z) = 1 - \frac{5}{6}z + \frac{1}{6}z^2 = \frac{1}{6}(z-2)(z-3)$ (roots 2, 3 outside the unit circle $\Rightarrow$ causal). Multiply the model by $X_{t-\tau}$ and take expectations. Since the process is causal, $X_{t-\tau}$ is a one-sided combination of $\varepsilon_{t-\tau}, \varepsilon_{t-\tau-1}, \dots$, so $\mathbb{E}[\varepsilon_t X_{t-\tau}] = 0$ for $\tau \geq 1$ and $\mathbb{E}[\varepsilon_t X_t] = \psi_0 \sigma^2 = \sigma^2$. Hence, for $\tau \geq 1$,

$$\gamma_\tau - \frac{5}{6}\gamma_{\tau-1} + \frac{1}{6}\gamma_{\tau-2} = \mathbb{E}[\varepsilon_t X_{t-\tau}] = 0 \quad \implies \quad \gamma_\tau = \frac{5}{6}\gamma_{\tau-1} - \frac{1}{6}\gamma_{\tau-2},$$

(using $\gamma_{-1} = \gamma_1$ when $\tau = 1$). The boundary relation at $\tau = 0$ keeps the noise term:

$$\gamma_0 - \frac{5}{6}\gamma_1 + \frac{1}{6}\gamma_2 = \mathbb{E}[\varepsilon_t X_t] = \sigma^2.$$

(b) The recurrence at $\tau = 1, 2$ (with $\gamma_{-1} = \gamma_1$) gives the autocorrelations via Yule-Walker:

$$\rho_1 = \frac{\phi_1}{1-\phi_2} = \frac{5/6}{1+1/6} = \frac{5/6}{7/6} = \frac{5}{7}, \quad \rho_2 = \phi_1\rho_1 + \phi_2 = \frac{5}{6} \cdot \frac{5}{7} - \frac{1}{6} = \frac{25}{42} - \frac{7}{42} = \frac{18}{42} = \frac{3}{7}.$$

Plug into the $\tau = 0$ relation, $\gamma_0(1 - \phi_1\rho_1 - \phi_2\rho_2) = \sigma^2$:

$$1 - \frac{5}{6} \cdot \frac{5}{7} - \left(-\frac{1}{6}\right) \frac{3}{7} = 1 - \frac{25}{42} + \frac{3}{42} = 1 - \frac{22}{42} = \frac{20}{42} = \frac{10}{21}, \quad \gamma_0 = \frac{21}{10}\sigma^2.$$

Therefore $\gamma_1 = \rho_1\gamma_0 = \frac{5}{7} \cdot \frac{21}{10}\sigma^2 = \frac{3}{2}\sigma^2$ and $\gamma_2 = \rho_2\gamma_0 = \frac{3}{7} \cdot \frac{21}{10}\sigma^2 = \frac{9}{10}\sigma^2$.

The homogeneous recurrence has characteristic equation Φ , whose roots correspond to the reciprocal roots $\frac{1}{2}, \frac{1}{3}$, so the general solution is $\gamma_\tau = c_1 \left(\frac{1}{2}\right)^\tau + c_2 \left(\frac{1}{3}\right)^\tau$. Match $\gamma_0 = \frac{21}{10}\sigma^2$ and $\gamma_1 = \frac{3}{2}\sigma^2$:

$$\begin{aligned} c_1 + c_2 &= \frac{21}{10}\sigma^2, \\ \frac{1}{2}c_1 + \frac{1}{3}c_2 &= \frac{3}{2}\sigma^2. \end{aligned}$$

Subtracting $\frac{1}{3} \times (\text{first})$ from the second, $\frac{1}{6}c_1 = \left(\frac{3}{2} - \frac{21}{30}\right)\sigma^2 = \frac{45-21}{30}\sigma^2 = \frac{24}{30}\sigma^2$, so $c_1 = \frac{24}{5}\sigma^2$ and $c_2 = \frac{21}{10}\sigma^2 - \frac{24}{5}\sigma^2 = -\frac{27}{10}\sigma^2$. Hence

$$\boxed{\gamma_\tau = \left[\frac{24}{5} \left(\frac{1}{2}\right)^\tau - \frac{27}{10} \left(\frac{1}{3}\right)^\tau \right] \sigma^2, \quad \tau \geq 0}$$

and $\gamma_{-\tau} = \gamma_\tau$. (Check: $\tau = 0 \rightarrow \frac{48-27}{10} = \frac{21}{10}$; $\tau = 1 \rightarrow \frac{24}{10} - \frac{27}{30} = \frac{72-27}{30} = \frac{45}{30} = \frac{3}{2}$; $\tau = 2 \rightarrow \frac{24}{20} - \frac{27}{90} = \frac{6}{5} - \frac{3}{10} = \frac{9}{10}$, all matching.)

Independent check (via ψ -weights): $\psi_j = 2\left(\frac{1}{2}\right)^j - \left(\frac{1}{3}\right)^j$ solves $\psi_j = \frac{5}{6}\psi_{j-1} - \frac{1}{6}\psi_{j-2}$ with $\psi_0 = 1, \psi_1 = \frac{5}{6}$, and $\sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$ reproduces $\frac{21}{10}, \frac{3}{2}, \frac{9}{10} \dots$ exactly.

Problem 4 (5 points)

(a) With $X_t = \varepsilon_t - \frac{1}{2}\varepsilon_{t-1}$, bilinearity and $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ give

$$\gamma_\tau = \begin{cases} (1 + \theta^2)\sigma^2 = \left(1 + \frac{1}{4}\right)\sigma^2 = \frac{5}{4}\sigma^2, & \tau = 0, \\ -\theta\sigma^2 = -\frac{1}{2}\sigma^2, & \tau = \pm 1, \\ 0, & |\tau| \geq 2, \end{cases} \quad \rho_1 = \frac{-\theta}{1 + \theta^2} = \frac{-1/2}{5/4} = -\frac{2}{5}.$$

(b) For a general MA(1), $\rho_1 = \frac{-\theta}{1 + \theta^2}$, so

$$|\rho_1| = \frac{|\theta|}{1 + \theta^2} \leq \frac{1}{2} \iff 2|\theta| \leq 1 + \theta^2 \iff 0 \leq \theta^2 - 2|\theta| + 1 = (|\theta| - 1)^2,$$

which always holds (equality iff $|\theta| = 1$). Equivalently, by AM-GM $|\theta| \leq \frac{1}{2}(1 + \theta^2)$. So $|\rho_1| \leq \frac{1}{2}$.

PACF behaviour: for an MA(1) the **PACF tails off** (decays geometrically in magnitude with alternating sign, never cutting to exactly 0), while it is the ACF that cuts off after lag 1. (This is the diagnostic mirror of the AR(1), whose PACF cuts off after lag 1 and ACF tails off.)

(c) **Invertible:** $\Theta(z) = 1 - \frac{1}{2}z$ has root $z = 1/\theta = 2$ with $|2| > 1$ (outside the unit circle), so $\{X_t\}$ is invertible.

A different MA(1) with the same ACVS is the *reciprocal-root twin*: take coefficient $\vartheta = 1/\theta = 2$ and white-noise variance $\sigma_\eta^2 = \theta^2\sigma^2 = \frac{1}{4}\sigma^2$, i.e.

$$Y_t = \eta_t - 2\eta_{t-1}, \quad \text{Var}(\eta_t) = \frac{1}{4}\sigma^2.$$

Its autocovariances are

$$\gamma_0^Y = (1 + \vartheta^2)\sigma_\eta^2 = (1 + 4) \cdot \frac{1}{4}\sigma^2 = \frac{5}{4}\sigma^2, \quad \gamma_{\pm 1}^Y = -\vartheta\sigma_\eta^2 = -2 \cdot \frac{1}{4}\sigma^2 = -\frac{1}{2}\sigma^2,$$

and $\gamma_\tau^Y = 0$ for $|\tau| \geq 2$ — *identical* to $\{X_t\}$ in part (a). So the two are indistinguishable from second-order structure alone (MA non-identifiability). The **invertible** representation is $\{X_t\}$, whose MA root $z = 2$ lies *outside* the unit circle; the twin $\{Y_t\}$ has root $z = \vartheta^{-1} = \frac{1}{2}$ *inside* the circle and is therefore *not* invertible. Imposing invertibility selects $\{X_t\}$ as the unique model.